

Announcements

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Weeks 1-4

- Introduction
- Agent- and individual-based simulation
- Fundamental algorithms for biological processes

Weeks 1-4

- Introduction

- Data-generating processes
- Models as algorithms (rather than equations)
- Equations as useful abstractions to algorithms
- Biological versus observation processes
- Modeling workflows

Weeks 1-4

- Agent- and individual-based simulation
 - Discrete events
 - Procedural programming
 - Object oriented programming and Python classes
 - Using high performance computing (HPC) resources

Weeks 1-4

- Fundamental algorithms for biological processes
 - Stochastic state-changes: birth, death, movement, consumption, growth, infection, mutation, recombination, branching
 - Distributions: exponential, binomial, Poisson, multinomial, etc
 - derived from biological and physical mechanisms
 - pseudo-random numbers
 - variance-mean relationships
 - Gillespie algorithm for continuous-time processes
 - rates and exponential waiting times for discrete events

Learning philosophy

- Algorithms first
 - models, data generating processes
 - understanding (nature, stats, etc)
 - getting stuff done (solving problems, automating)
- Other stuff is housekeeping
 - data structures, data types, libraries

Today

- Review: models as algorithms, data-generating processes
- Biological vs observation process
- Modeling workflow
- Pseudorandom numbers (algorithms)
- Uniform, Bernoulli, binomial distributions
- Intro to stochastic processes
- Intro to agent-based simulation

Models as algorithms

- "What happens next?" thinking
- First this happens
- then that happens
- = algorithm

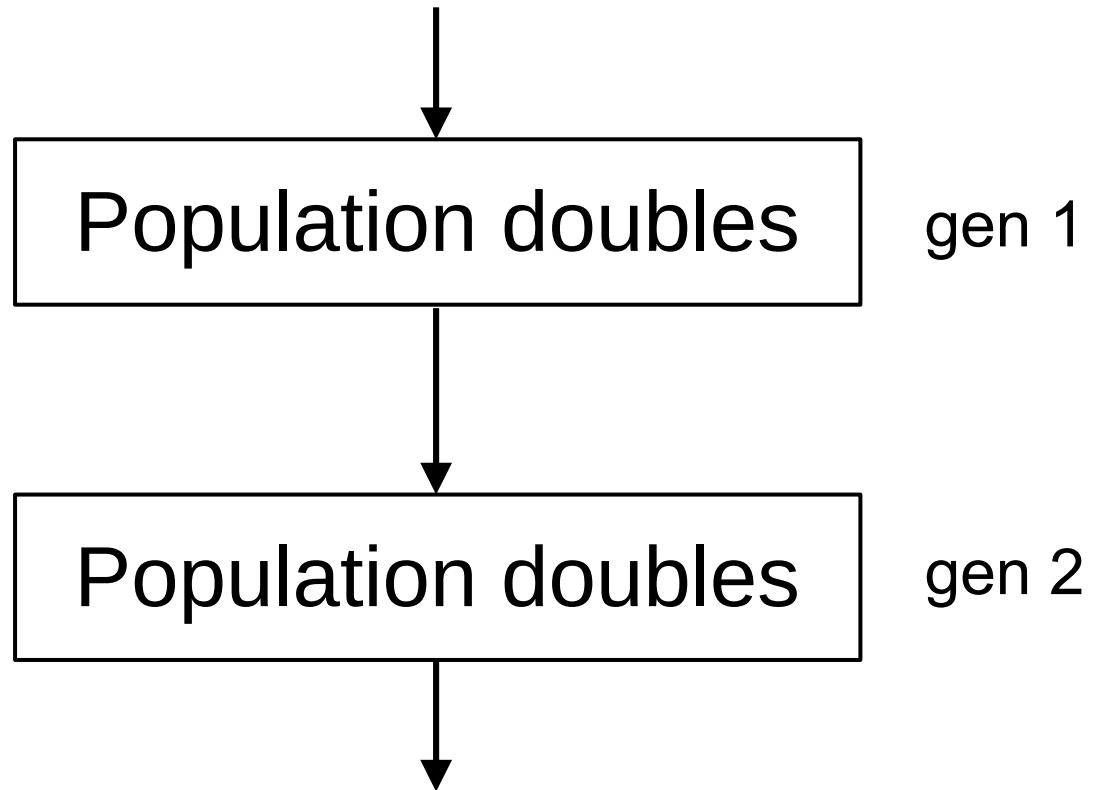
Key message:

Biological processes tend to be algorithmic

What happens next?

Event sequence diagram

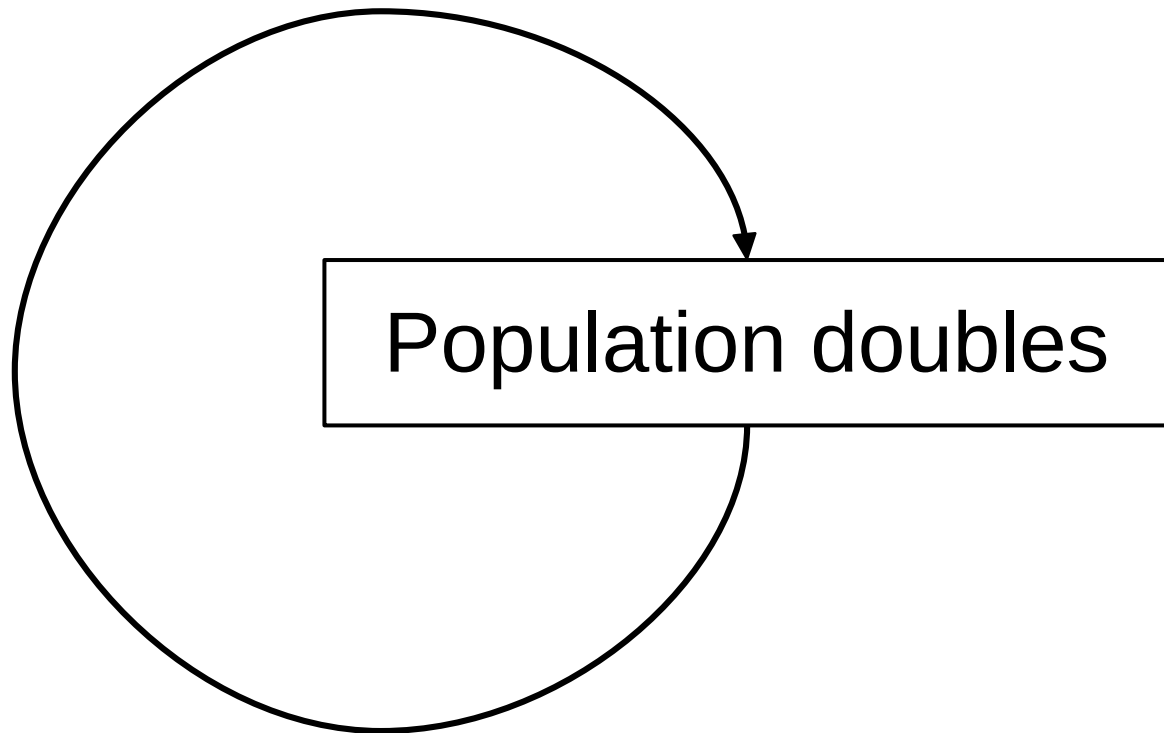
e.g. population
doubles each
generation



What happens next?

Event sequence diagram

simplified



Art of modeling

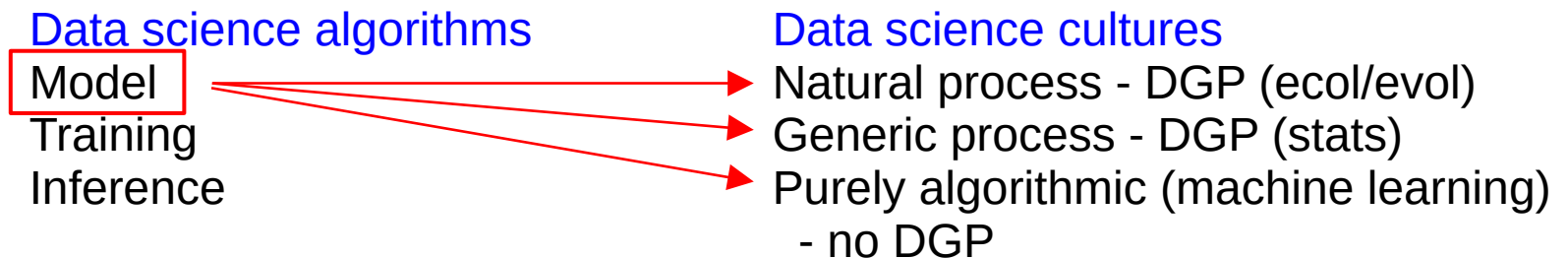
- How much detail?
 - abstract, simplify, scales
 - assumptions
- e.g. doubling population
 - skipping over births, deaths, processes per individual, physiology etc
 - timescale = generation
 - biological scale = population

Data generating process(es)

- Where do data come from?
- An **actual physical process** involving fundamental particles of the universe
- Includes
 - **ecological/evolutionary** process(es)
 - **observing** process(es)

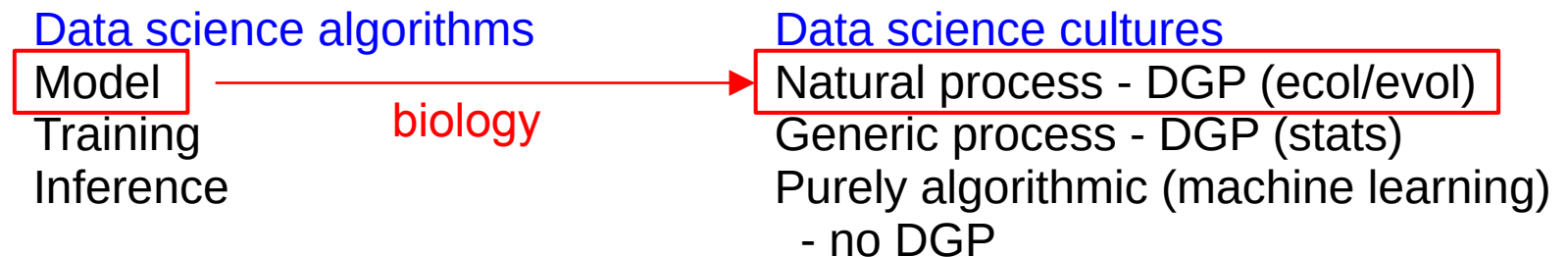
Data generating process

- Key to scientific understanding
 - How does the system work?
- How to **model** the DGP?



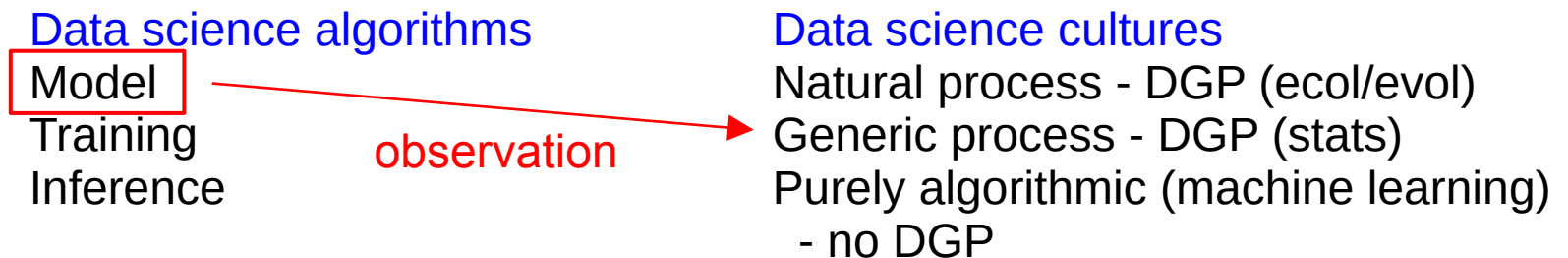
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Data generating process

- Key to scientific understanding
 - How does the system work?
- How to **model** the DGP?



Stochastic processes

- **Substitute** for all the stuff we don't know
- **Uncertainty** about finer-scale processes
- Is the world deterministic or stochastic?
 - my view: **depends on scale**
 - **individual scale** is stochastic
 - individuals **perceive** the world as uncertain

Stochastic processes

- Start learning this topic today
- Review probability along the way
 - from an algorithms perspective
- Distributions are key
 - most arise from fundamental processes
 - algorithmic derivations

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Stochastic processes

- Fundamental idea:
- If you do it again the outcome will be different
- Each realization is drawn from a fundamental process that is nevertheless the same

Stochastic processes

- Random variable
- X (uppercase) is a random variable
 - members of X share the same fundamental process
- x (lowercase) is a particular outcome
 - one realization of the fundamental process
 - data (could be a vector etc)

Stochastic processes

- What is a distribution?
- Describes the probability density (continuous) or probability mass (discrete) of a random variable
- Representation of the fundamental process
- Summary of the fundamental attributes of the random variable

Uniform distribution

- “Completely random” random variable
- **Continuous** real numbers between a and b
- All the numbers have **equal probability**

$$f(x; a, b) = \begin{cases} \frac{1}{(b-a)} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad \begin{array}{l} \text{Probability} \\ \text{density} \\ \text{function} \\ \text{(pdf)} \end{array}$$

$X \sim \text{Uniform}(a, b)$ “X is distributed as ...”

Probability density function

- Probability is the **area** under the probability density function
- Total probability = 1

Uniform(0,1) distribution

$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{pdf}$$

$$X \sim \text{Uniform}(0,1)$$

Foundational

Completely random numbers between 0 and 1

Generating random numbers

- How does the computer do it?
- Pseudo-random numbers
 - **deterministic** sequence that “looks” random
 - pass various tests for randomness

Generate Uniform(0,1)

- How do we generate a uniform random number?
- Arbitrary algorithms found sort of by accident, e.g. Lewis et al 1969
 - iterative function

$$I_{j+1} = (a I_j) \bmod m$$

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$$a = 7^5 = 16807, \quad m = 2^{31} - 1 = 2147483647$$

Modern algorithms

- 1 generate random bits
- 2 turn those into integers
- 3 divide by largest integer to give $\text{Uniform}(0,1)$

Modern algorithms

- **R:** Mersenne_Twister
- **Numpy:** PCG64 (Permuted Congruential Generator)

Bernoulli distribution

- Probability of an event (discrete)
- $\Pr(X=1)=p$
- $\Pr(X=0)=1-p$

$$f(x; p) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Probability
mass
function
(pmf)

$$X \sim \text{Bernoulli}(p)$$